Vinith Kuruppu

vinithkuruppu5@gmail.com | vink23.github.io | linkedin.com/in/vinithkuruppu | 815-922-3560

Education

University of California, Berkeley – Master of Science in Data Science (GPA: 3.97/4.0) Dec 2025

- **Honors:** Hal R. Varian Award – Distinguished Capstone Project
- **Relevant Coursework:** Machine Learning, Natural Language Processing, Computer Vision, Generative AI, Research Design and Application, Data Engineering, Experiments and Causal Inference

Loyola University Chicago – Bachelor of Science in Computational Neuroscience May 2022

Skills & Technologies

Languages: Python, SQL, R, Bash
Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, QLoRA/LoRA (PEFT), RAG, FAISS, LangChain, Vision Transformers, CNNs, ASR, Large Language Models (LLMs)
Analytics & Experimentation: Experimental Design, A/B Testing, Causal Inference, Statistical Modeling, Hypothesis Testing, Power Analysis, Fairness Analysis
Systems & Data: AWS (SageMaker, EC2, S3), Docker, Spark/PySpark, PostgreSQL, Git, MLflow, Weights & Biases, GCP, MongoDB, Pandas, NumPy, CI/CD

Experience

Machine Learning Engineer, University of California, Berkeley | Remote Aug 2025 - Present

- Engineer an LLM-powered clinical decision-support system for physician-facing Q&A, combining dense retrieval, cross-encoder reranking, and grounded generation over medical literature to produce faithfully sourced diagnostic answers.
- Build a reproducible evaluation pipeline to benchmark 54 retrieval, reranking, and generation configurations across recall, clinical relevance, and answer grounding, replacing ad hoc model selection with a systematic, versioned process.
- Identify failure patterns across disease categories and patient age groups through stratified error analysis, directly informing retrieval and model changes that reduced performance gaps across underrepresented diagnostic categories.

Graduate Teaching Instructor, University of California, Berkeley | Remote Jan 2025 - Present

- Design and grade assignments for 80+ students in causal inference, experimental design, and statistical analysis, providing written feedback to reinforce applied research skills.

AI/ML Research Associate, RAND Corporation | Remote May 2024 - May 2025

- Developed fairness-aware allocation models that informed the allocation of $70 billion across 3,300+ hospitals, reducing allocation error and improving equity in disaster reimbursement across U.S. regions.
- Shipped an Excel-integrated classification tool backed by an AWS ML endpoint, enabling non-technical analysts to label 1M+ expense records with one click; improved accuracy by 22% vs. a manual baseline and was adopted in FEMA allocation review workflows.
- Designed simulation frameworks for disaster-response scenarios under uncertainty and ran sensitivity analyses across 50+ parameter configurations; findings directly shaped FEMA preparedness protocols.

Data Scientist, Exponent Incorporated | Chicago, IL Jul 2022 - May 2024

- Engineered data pipelines processing high-frequency physiological signals from 3,000+ Apple Watch devices, reducing integration time by 15% and enabling rapid iteration on Apple Health's hypertension detection algorithms.
- Led end-to-end FDA-compliant validation for biosensor clinical trials, achieving a 100% pass rate on QA audits through rigorous statistical validation and bias analysis; work enabled product launch for a health monitoring platform.

- Conducted fairness-aware analysis of biosensor signals across demographic subgroups (age, gender, skin tone), identifying algorithm performance disparities that informed design changes, improving signal-quality acceptance by 25% and reducing group-level performance gaps.

Computer Vision Research Assistant, Loyola University Chicago | Chicago, IL Jan 2020 - Jun 2022

- Led a 6-person team developing bio-inspired CNN architectures with attention mechanisms to model human visual perception, establishing lab benchmarks for human behavioral data from EEG/eye-tracking studies.
- Improved model accuracy by 11% vs. prior lab baseline via targeted augmentation, systematic hyperparameter optimization, and architecture refinements on object recognition tasks.
- Engineered Python data pipelines for image and behavioral datasets, reducing preprocessing time by 30% through parallelization and automated quality checks.

Projects

SurgiRAG: Retrieval-Augmented Generation for Surgical Question Answering 2025

PyTorch, LLaMA-3.2-11B (QLoRA), BioBERT, BGE, FAISS, Hugging Face

- Built a domain-adaptive RAG pipeline for Q&A by fine-tuning LLaMA-3.2-11B with LoRA and pairing BioBERT retrieval with cross-encoder reranking, improving LLM-judged faithfulness from 0.25 to 0.66 vs. the base model.
- Designed a custom LLM-as-judge evaluation framework using GPT-4o-mini to assess faithfulness and fluency tradeoffs not well captured by standard metrics like ROUGE and BLEU.
- Evaluated 12 retrieval, reranking, and generation configurations, identifying cross-encoder reranking as the highest-leverage component for improving answer quality.

AgeVoicE: Adapting Whisper for Disfluent Speech (Hal R. Varian Award Winner) 2025

PyTorch, Whisper, LoRA, Hugging Face, AWS

- Fine-tuned Whisper ASR models using LoRA for older-adult and dementia-affected speech, improving WER by 17% vs. baseline; deployed via Hugging Face Inference API and an interactive web demo for real-time transcription.
- Built an end-to-end AWS training pipeline with distributed training, custom preprocessing (noise removal, disfluency handling, audio normalization), experiment tracking, and systematic hyperparameter sweeps (LR, LoRA rank, dropout).
- Performed stratified failure analysis across acoustic conditions and speaker characteristics using regression modeling and hypothesis testing to identify drivers of elevated WER; findings guided targeted data augmentation to mitigate demographic bias.

PathoVision: Hybrid Vision Transformer for Brain Tumor Classification 2025

PyTorch, DINOv2 (Vision Transformer), scikit-learn, OpenCV

- Built a brain tumor MRI classifier achieving 96.7% test accuracy by combining DINOv2 embeddings with handcrafted edge features, outperforming CNN baselines with a fraction of the compute.
- Identified contrast and orientation artifacts as key misclassification drivers through cross-source error analysis, leading to preprocessing changes that improved robustness across imaging sources.